Arunabha Majumder

Research Engineer | Wearable Robotics, Exoskeletons, Actuators & Experimental Validation

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Robot Design Lab, Division of Mechatronics
KTH Royal Institute of Technology, Stockholm, Sweden

Personal Letter: Research Engineer in Robot Design and Mechatronics (REMA project)

Dear Members of the Hiring Committee,

I am applying for the position of Research Engineer in robot design and mechatronics with a focus on soft and wearable robotics. I hold an MTech in Mechatronics and a BTech in Mechanical Engineering, and I am completing my PhD in Mechatronics and Robotics at Aalborg University, where I design and test exoskeletons. The REMA project's goal, exoskeletons that help older adults stay mobile, is the kind of work I want to do next.

Wearable robotics has been the thread through my whole education. My Bachelor's capstone was an upper-limb rehabilitation exoskeleton, which I designed, machined, assembled and tested; it won the Best Major Project Award and a TATA Technologies Innovation Grant. My Master's thesis at CSIR-CMERI was on soft pneumatic artificial muscles. I fabricated and assembled the muscles to an existing lab design, collected test data on the lab's setup and fed it into a neural network developed in the group, then used its output to design the outer PID loop of a gain-scheduled controller. Position error came down to 0.3-0.78% (IEEE CONECCT 2022). That project is where I first saw how hard and how interesting soft actuators are to control.

In my PhD project VIEXO, I designed a hybrid-actuated shoulder exoskeleton that combines a compliant parallel spring with a small 6 Nm motor. I carried out the mechanical design in SolidWorks, built the prototypes through 3D printing, and integrated the motor, encoder, force sensors and embedded control. The design reduced energy consumption by more than 25% and received the Best Research Paper Award at IFToMM ISRM 2026. With two colleagues I worked on an Assist-as-Needed controller, where I helped with the equations such as the Fourier fit and with how the GUI should work. It reduced user muscle effort by more than 15%. On a 5-bar planar parallel robot I developed a force-estimation method using variable-stiffness actuators and the robot Jacobian, validated against a force sensor to below 10% RMS error and used to measure human arm impedance.

Experimental work with people is central to wearable robotics, and I have conducted experiments with 12 participants using EMG and motion capture across several assistance modes and load conditions. I have also supervised Bachelor's students and taught university courses, which matches the student supervision part of this role.

What draws me to Robot Design Lab is the combination the REMA project brings together: soft robotics, sensing, actuation, adaptive control and user-centred design. My experience covers rigid and compliant exoskeleton actuation and hands-on work with soft pneumatic muscles, and I would like to go deeper into soft wearable structures, to use multibody simulation such as Simscape Multibody more systematically, and to design with older adults' needs in mind. I would bring a hands-on, iterative way of working: I sketch ideas by hand first (I hold a Diploma in Fine Arts), build functional prototypes quickly, test them and write up what we learn.

I would be glad to contribute to the REMA project and to the research of Robot Design Lab. Thank you for considering my application.

Kind regards,

Arunabha Majumder